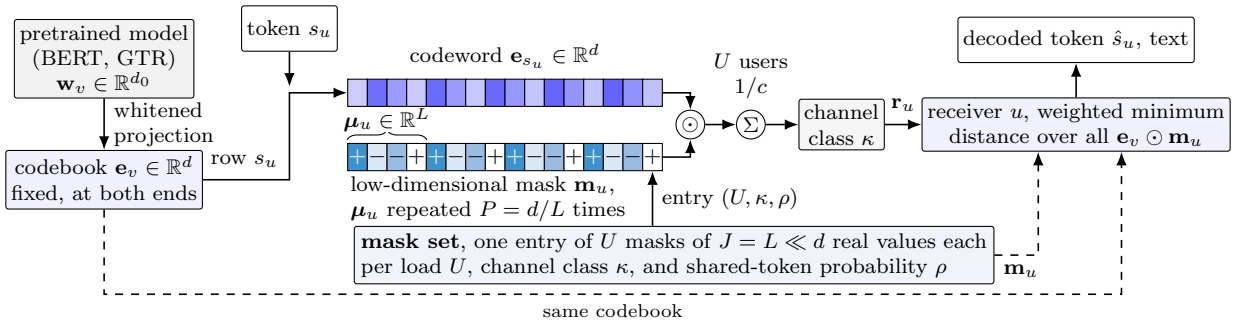

pretrained model (BERT, GTR) $\mathbf{w}_v \in \mathbb{R}^{d_0}$
whitened projection
codebook $\mathbf{e}_v \in \mathbb{R}^d$ fixed, at both ends
row s_u
token s_u
codeword $\mathbf{e}_{s_u} \in \mathbb{R}^d$
$\boldsymbol{\mu}_u \in \mathbb{R}^L$
low-dimensional mask $\mathbf{m}_u$, $\boldsymbol{\mu}_u$ repeated $P = d/L$ times
U users
$1/c$
channel class κ
$\mathbf{r}_u$
receiver u, weighted minimum distance over all $\mathbf{e}_v \odot \mathbf{m}_u$
decoded token $\hat{s}_u$, text
entry (U, κ, ρ)
mask set, one entry of U masks of $J = L \ll d$ real values each per load U, channel class κ, and shared-token probability ρ
$\mathbf{m}_u$
same codebook